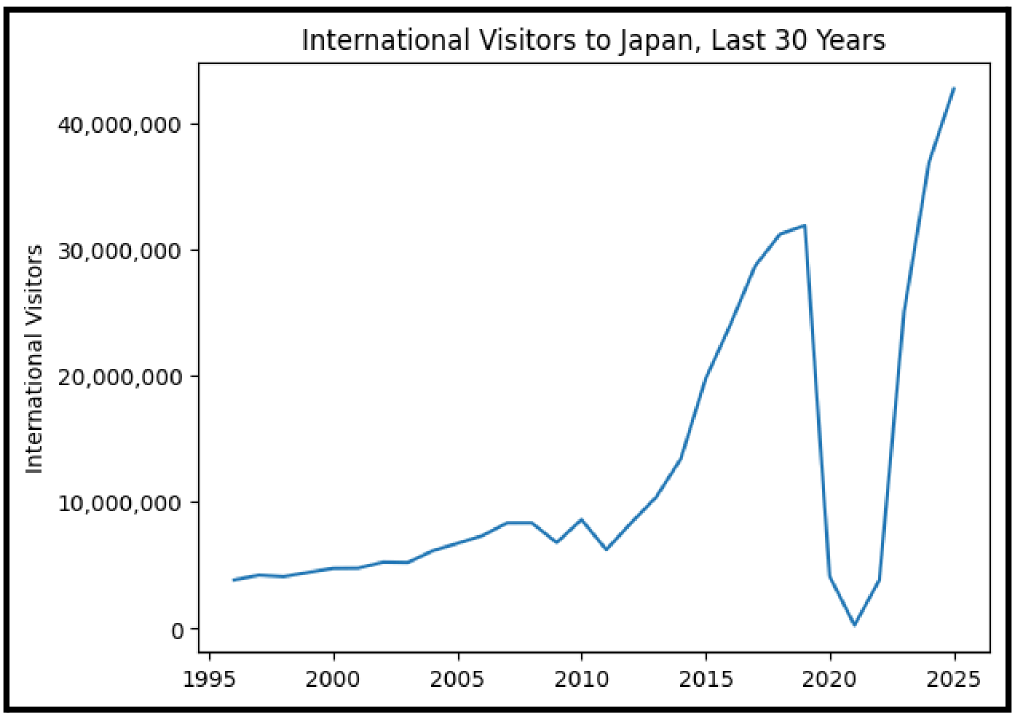
Joe Japan Tourism Analysis
June 7, 2026

Intro

For this project, I examined monthly international tourist volumes into Japan. In this part, I downloaded official data from the Japanese government and performed exploratory data analysis on it. I examined both long-term and seasonal trends in the data in an attempt to find the least crowded time to visit.

1. Long-Term Trends

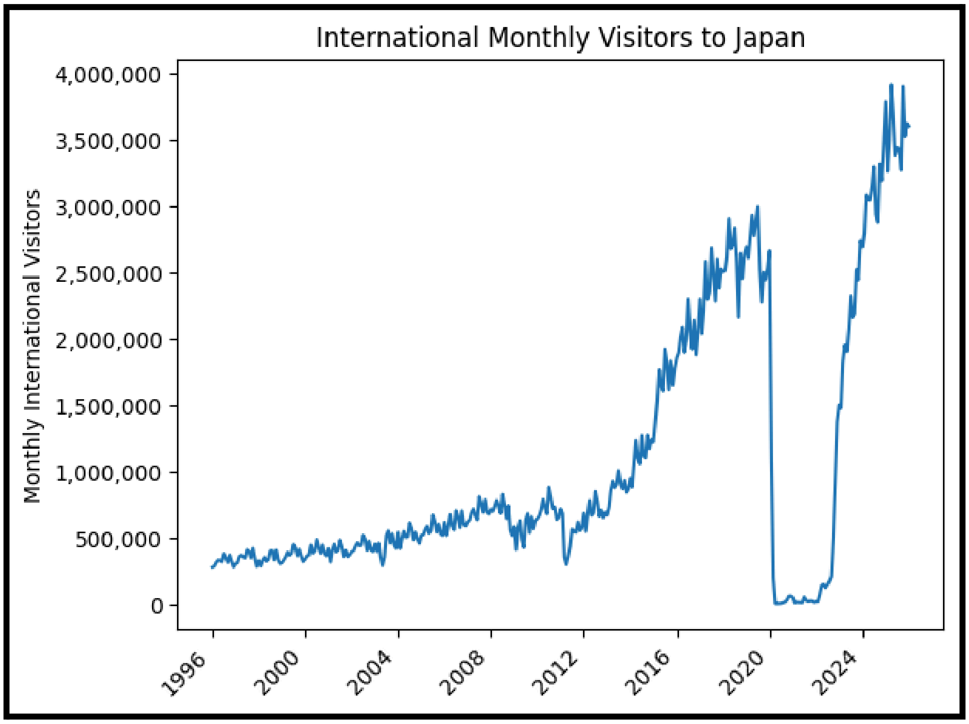
Even as Japan's resident population decreases, it can feel more crowded than ever as a tourist. That's probably because international visitor levels to the country are higher than ever.



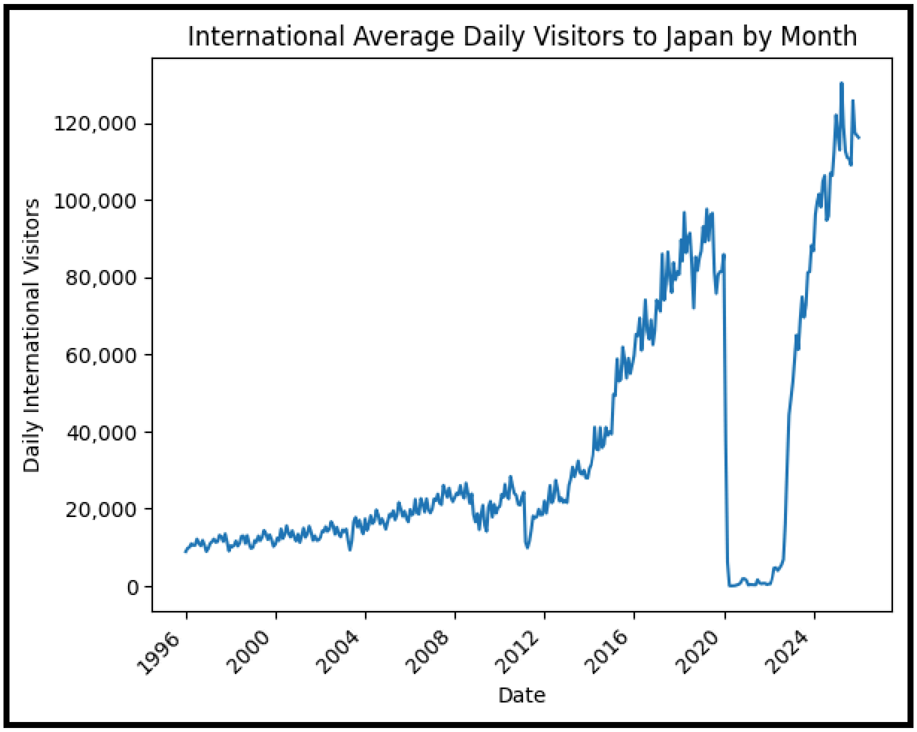
Note: All visitor data is taken from the www.tourism.jp website, specifically from the time series data downloadable at this website: https://www.tourism.jp/en/tourism-database/stats/inbound/

In a span of thirty years, the number of visitors to Japan increased more than ten-fold, from 3.8 million in 1996 to 42.7 million in 2025. As a semi-frequent visitor to the country, I (hypocritically) want to visit when fewer other visitors do. That's why I set about using Japan's official visitor in-flow statistics to plan the timing of my next trip.

JTB Tourism Research & Consulting Co provides international visitor data dating back to 1996. The data is provided on a monthly basis, which is arguably not granular enough in the context of trip planning. It also does not account for domestic tourism volumes. However, I believe this data can still prove instructive as a general measure of “crowdedness” over a broad period.



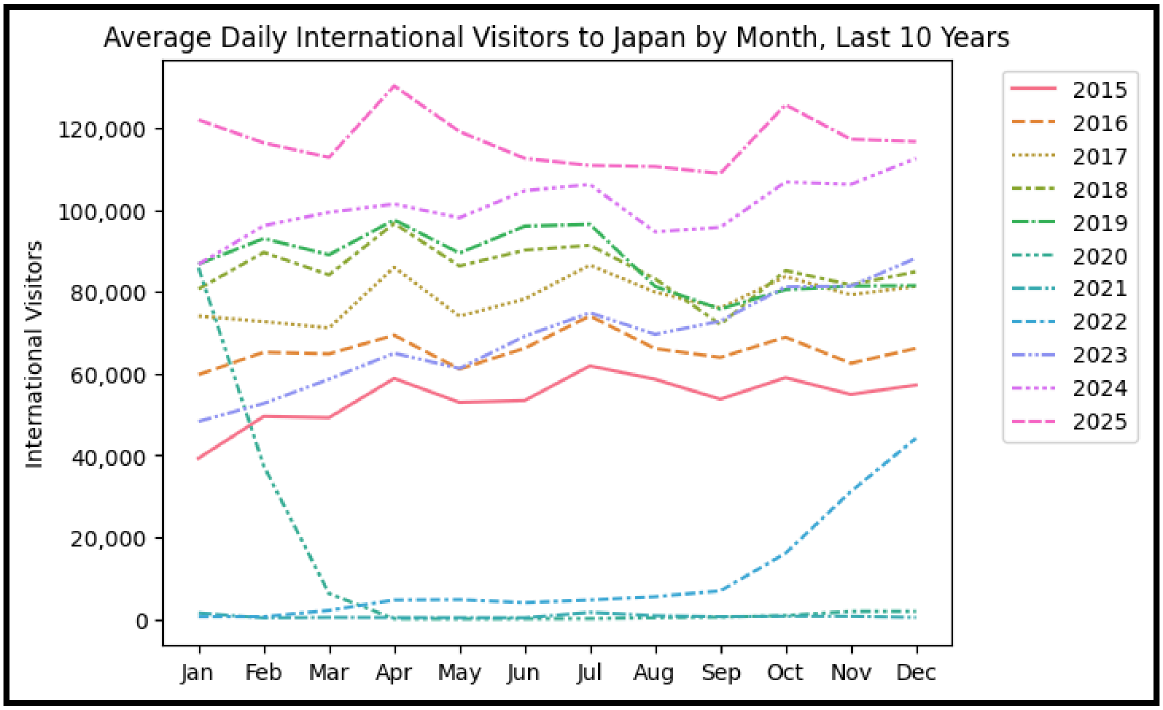
Here is the data broken down by month. This does not provide the best measure of “crowdedness” because months have a variable number of days. A normalized graph of average daily visitors would be more instructive.



The COVID pandemic, during which Japan heavily restricted incoming visitor levels, looms large in the dataset. Although it decimated visitor levels from ~2020-2022, crowds appear to have reached record highs since. It's an important reminder of the capability of outlier or "black swan" events to massively disrupt global tourism. However, the distorting effects of COVID visitation should be kept in mind when using past data to predict visitation levels during more "normal" periods of international travel.

2. Seasonal Trends

To what degree does the season influence visitor volumes? Below is a graph showing monthly visitor volumes stratified by year.



It's difficult to make out the seasonal trends because it seems as if the difference in arrivals across years is much bigger than the difference within years. An aggregated measure of seasonal trends over the whole dataset would be more insightful.

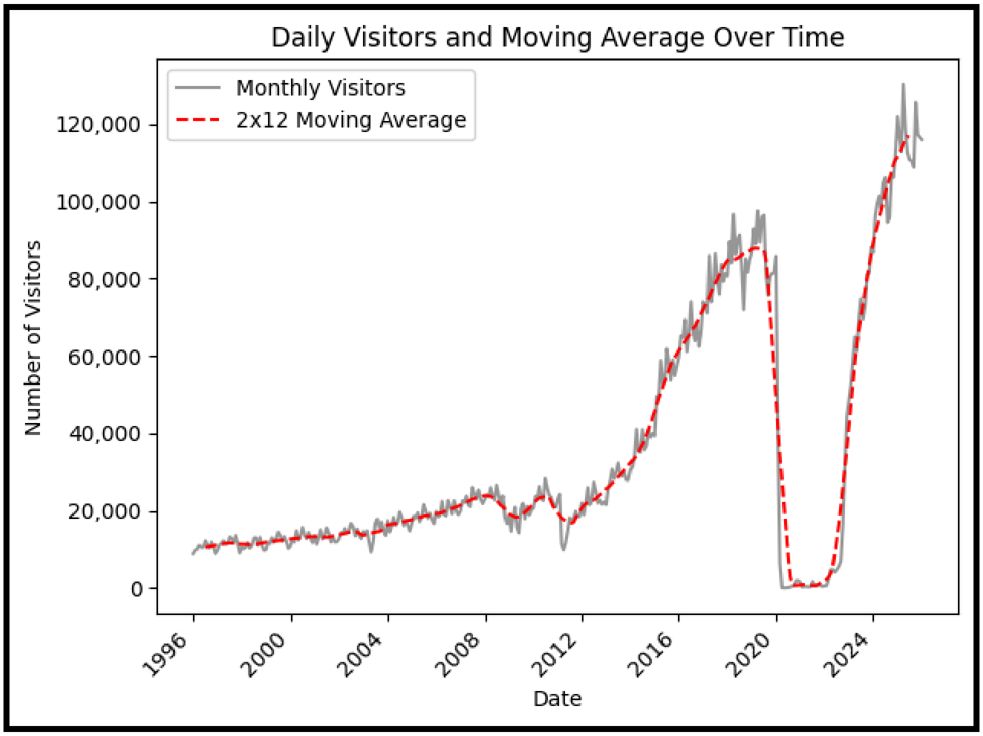
Time series data such as this can be modeled as being composed of the following three components:

- T - the trend-cycle component. This captures the overall trend of the data.
- S - the seasonal component. This captures the variation due to seasonality. For this dataset, that would be the variation during one 12 month cycle.
- R - the remainder component. This captures unexplainable noise in the data

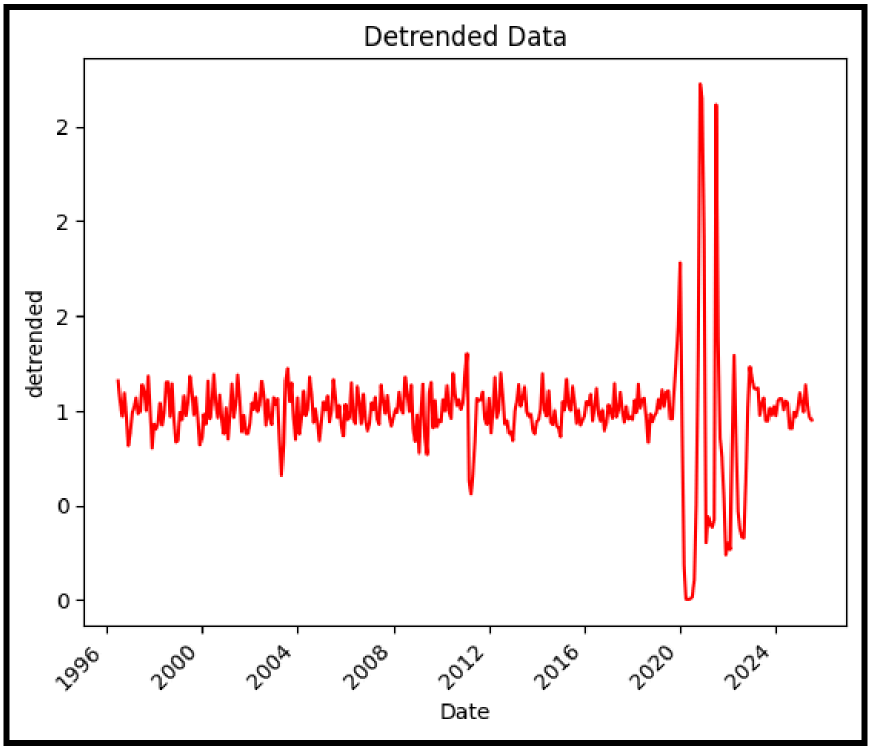
In this case, because the seasonal variation appears to increase in proportion with the overall trend, it makes sense to model the visitation levels as the product of the three components.

$$y_t = T_t \times S_t \times R_t$$

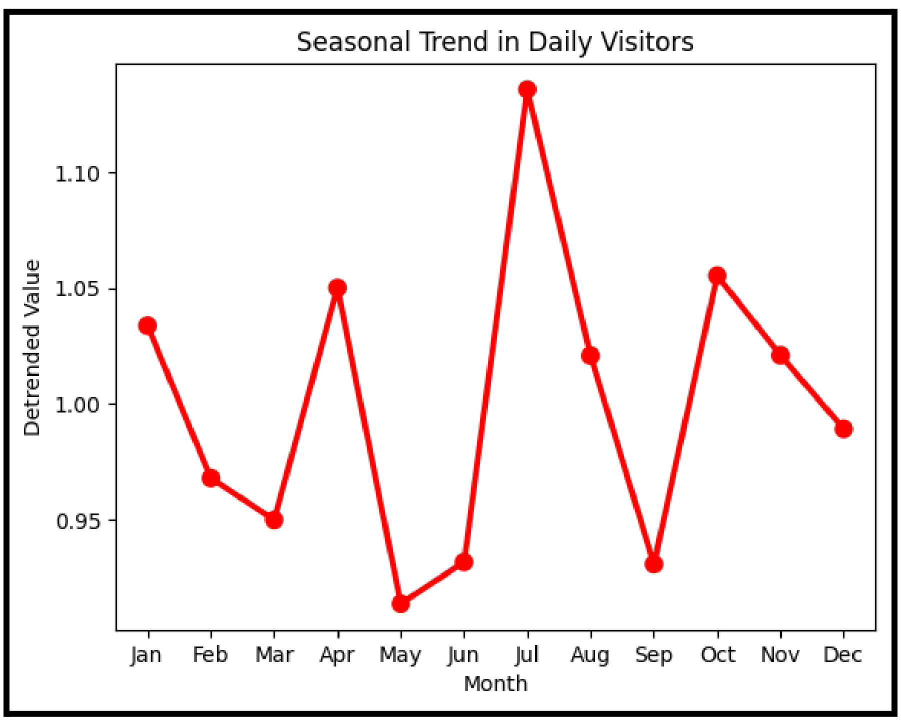
The trend-cycle component can be estimated using a moving average. A moving average of the order 2 x 12 is best for this data, since we want the moving average to be computed symmetrically over the course of one cycle, in this case, a year.



The moving average starts and ends six months away from the end points of the data. That is because this moving average incorporates the six months on each side of the data point, making values of moving averages closer than six months away from the endpoints impossible to calculate. The original data can be divided by the moving average to get an estimate of the de-trended data.



Averaging over these values for each month gives an estimate for the seasonal component of the tourist data:



This graph shows that historically, July has been the biggest month for tourist arrivals. July has an average seasonal value of ~1.14, meaning that historically arrivals to the country were ~14% greater in July relative to the baseline. May has the lowest seasonal trend at ~0.91, meaning the month experiences 9% lower visitation on average.

Part 2

In the next segment, I will deploy forecasting methods in a more systematic way to predict future visitation arrivals.

Sources

- Equations and theory
  - Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice (3rd ed.)*. OTexts. https://otexts.com/fpp3/
- Japan Tourism data
  - Japan Tourism Marketing Co. (2026). Statistics of Overseas Residents' Visits to Japan (Historical figures/by country/by purpose) [Data set]. https://www.tourism.jp/wp/wp-content/uploads/2026/03/JTM_inbound_20260304eng.xlsx